Electrical Safety Analyzer



Category: **Electrical & Electronic Test Instruments** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Electrical safety analyzer

What it is used for

An electrical safety analyzer tests electrical safety characteristics of equipment, such as protective earth continuity, insulation resistance, leakage current, and selected functional checks, depending on the model. Correct test selection, safe connections, and accurate recording help identify electrical hazards before equipment is returned to service.

Main parts

- Analyzer main unit and display
- Test leads and probes
- Protective earth continuity test connections
- Insulation resistance and leakage current test connections, where fitted
- Equipment-under-test power outlet or connection
- Test controls, adapters, and calibration label

Before you start

1. Only trained and authorised personnel should operate the analyzer and interpret its test results.
2. Identify the equipment under test (EUT), its rated voltage, protection class, and applicable test procedure.
3. Read the analyzer manual and the EUT manufacturer's instructions to confirm which tests are appropriate.
4. Inspect the analyzer, leads, probes, adapters, and connectors for damage, exposed conductors, or contamination.
5. Verify that the analyzer is within its calibration period and that the correct test limits and settings are selected.
6. Remove the EUT from normal service and disconnect it from other equipment unless the approved test procedure requires a specific connection.

How to use

1. Place the analyzer on a stable, dry surface in a controlled test area.
2. Switch on the analyzer and perform its specified self-test or readiness check.
3. Connect the test leads and EUT according to the manufacturer's connection diagram and the selected test procedure.
4. Confirm that the EUT is correctly identified and that its normal operating controls are set as required for the test.

5. Perform protective earth continuity testing where applicable, using the specified test current and acceptance limit.
6. Perform insulation resistance or leakage current testing only when appropriate for the EUT and permitted by its instructions.
7. Select the correct test configuration for the EUT's protection class, applied parts, and intended use.
8. Start each test using the analyzer's approved sequence and keep personnel clear of exposed conductive parts during energised tests.
9. Observe displayed readings, warnings, and test status; stop if a connection is loose, a fault appears, or the setup becomes unsafe.
10. Compare results with the applicable manufacturer's limits, facility procedures, and relevant standards for the equipment type.
11. If a test fails, label the EUT as failed or not for use, document the result, and refer it for qualified investigation.
12. After testing, follow the analyzer procedure to discharge any stored electrical energy and return outputs to a safe state.
13. Switch off the analyzer before disconnecting leads and remove connections carefully.
14. Record equipment identification, test selection, measured results, acceptance limits, instrument identification, and test date.

⚠ Safety

- Use the analyzer only within its rated voltage, current, and measurement category.
- Follow the analyzer manual because some tests apply hazardous voltage or current to the EUT.
- Never touch exposed conductors or the EUT during an energised test.
- Do not perform insulation resistance or high-voltage tests on sensitive electronics unless the equipment manufacturer and approved procedure permit them.
- Verify leads, adapters, protective earth connections, and calibration status before testing.
- Do not bypass interlocks or ignore warning indicators; stop if the analyzer or EUT behaves abnormally.
- Use appropriate electrical PPE and follow lockout/tagout requirements where applicable.

Care & storage

- Inspect and clean the analyzer, leads, probes, and adapters after use.
- Store leads without sharp bends and protect connectors from impact and moisture.
- Check calibration status and arrange calibration at the required interval or after suspected damage.
- Keep test records and note instrument identification and any failed tests.
- Remove damaged accessories from service and arrange qualified repair when required.

Common mistakes to avoid

- Selecting a test or limit that is unsuitable for the EUT.
- Touching the EUT or exposed conductive parts during an energised test.

- Ignoring calibration status or damaged test leads.
- Returning failed equipment to service without documented assessment and corrective action.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.